
```

function predictie_populatie_sua()
    ani_t = [1900; 1910; 1920; 1930; 1940; 1950; 1960; 1970; 1980; 1990;
2000; 2010; 2020];
    pop_y = [75.995; 91.972; 105.710; 123.200; 131.670; 150.700; ...
179.320; 203.210; 226.510; 249.630; 281.420; 308.790; 350.686];

    % translatez timpul pentru a evita erorile de rotunjire
    t_ref = ani_t - 1900;

    t_1975 = 1975 - 1900;
    t_2015 = 2015 - 1900;

    % modelul polinomial
    Matrice_Poli = [ones(size(t_ref)), t_ref, t_ref.^2, t_ref.^3];
    coef_poli = Matrice_Poli \ pop_y;

    eval_poli = @(t) coef_poli(1) + coef_poli(2)*t + coef_poli(3)*t.^2 +
coef_poli(4)*t.^3;
    pop_poli_1975 = eval_poli(t_1975);
    pop_poli_2015 = eval_poli(t_2015);

    % modelul exponential
    Y_log = log(pop_y);
    Matrice_Exp = [ones(size(t_ref)), t_ref];
    coef_exp_lin = Matrice_Exp \ Y_log;

    K = exp(coef_exp_lin(1));
    lambda = coef_exp_lin(2);

    eval_exp = @(t) K * exp(lambda * t);
    pop_exp_1975 = eval_exp(t_1975);
    pop_exp_2015 = eval_exp(t_2015);

    fprintf('modelul polinomial\n');
    fprintf('predictie populatie 1975: %.3f milioane\n', pop_poli_1975);
    fprintf('predictie populatie 2015: %.3f milioane\n\n', pop_poli_2015);

    fprintf('modelul exponential\n');
    fprintf('parametrii: K = %.3f, lambda = %.4f\n', K, lambda);
    fprintf('predictie populatie 1975: %.3f milioane\n', pop_exp_1975);
    fprintf('predictie populatie 2015: %.3f milioane\n', pop_exp_2015);
end

modelul polinomial
predictie populatie 1975: 211.485 milioane
predictie populatie 2015: 330.290 milioane

modelul exponential
parametrii: K = 81.411, lambda = 0.0124
predictie populatie 1975: 207.037 milioane
predictie populatie 2015: 340.598 milioane

```

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